

Data Mining Project – Milestone 3: Association, Clustering and Classification Tasks (June 2026)

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Abstract—This project presents a comprehensive data mining analysis of a global billionaires dataset containing 2,600 records and 35 variables collected from reliable sources such as Forbes, Bloomberg, and official economic databases. Through Exploratory Data Analysis (EDA), significant patterns in global wealth distribution were identified, including the dominance of the United States and China, the prevalence of self-made billionaires, and the concentration of wealth in sectors such as Finance & Investments, Manufacturing, and Technology.

A rigorous data preprocessing pipeline was applied, including missing value imputation, feature selection, industry and source standardization, feature engineering, discretization, binarization, and country-level aggregation. These transformations enabled the application of multiple data mining tasks.

Association rule mining uncovered relationships between billionaire characteristics and country-level socioeconomic indicators such as GDP, education, and life expectancy. Clustering techniques revealed distinct national wealth ecosystems, including Innovation Ecosystems, Rentier Paradises, and Extractive Economies, characterized by differences in sector composition, demographic structure, and economic conditions. Additionally, classification models, including Neural Networks, achieved an accuracy above 94% in predicting whether a billionaire is self-made.

The findings provide valuable insights into the structural factors associated with extreme wealth generation and their relationship with broader national economic models.

Index Terms—Data Mining, Exploratory Data Analysis (EDA), Data Preprocessing, Data Cleaning, Feature Engineering, Dimensionality Reduction, Association, Clustering, Classification, Artificial Neural Networks, Model Training.

I. INTRODUCTION

Global extreme wealth is a key structural phenomenon that shapes national economies and the social landscape. This project addresses the need to understand the underlying drivers of wealth concentration through a comprehensive data mining analysis. To this end, a global dataset of 2,600 records and 35 variables, collected from reliable sources such as Forbes and Bloomberg, was analyzed, combining individual billionaire attributes with country-level socioeconomic indicators.

The methodology focused on a rigorous Data Preparation process (including cleaning, feature engineering, and discretization) followed by advanced data mining tasks. Exploratory Data Analysis (EDA) identified key patterns, such as the prevalence of self-made billionaires and sectoral concentration in Finance, Manufacturing, and Technology, geographically dominated by the United States and China. The Association task uncovered structural links, such as the correlation between low industrial dependence and macroeconomic and social stability (inflation and life expectancy). The Clustering task allowed for the definition of national wealth archetypes, revealing three distinctive ecosystems: Innovation Ecosystem, Rentier Paradise, and Extraction Economy. Finally, the Classification task, using advanced models like Neural Networks, predicted with high accuracy (up to 94%) whether a billionaire had made their own fortune, highlighting the associated factors.

Collectively, the results of this analysis provide actionable perspectives on the structural drivers of global extreme wealth and their relationship with national economic models.

II. EXPLORATORY ANALYSIS

A) Dataset description

The dataset used in this study contains information about billionaires worldwide, collected from multiple reliable sources including Forbes, Bloomberg, government databases, financial institutions, and World Bank indicators. The data combines individual-level information, such as age, gender, source of wealth, industry, and citizenship, with country-level socioeconomic indicators including GDP, education enrollment, life expectancy, taxation, and population. This combination allows the analysis of both personal and macroeconomic factors associated with wealth creation.

The dataset was compiled through a combination of manual data collection and automated extraction techniques, including web scraping and APIs. This process resulted in a structured dataset suitable for exploratory analysis and data mining tasks. The final dataset contains 2,600 observations and 35 variables.

The variables can be grouped into four main categories: demographic attributes (e.g., age and gender), wealth-related variables (e.g., net worth, source of wealth, and industry), geographic information (e.g., country and city), and country-level socioeconomic indicators (e.g., GDP, education enrollment, inflation, and life expectancy). This diversity of variables provides a comprehensive view of the characteristics of billionaires and the economic environments in which they operate.

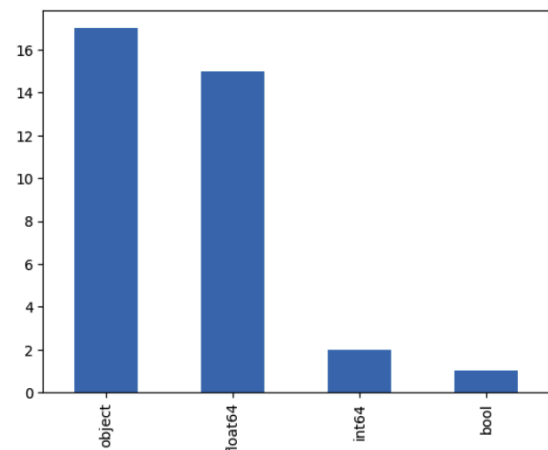


Figure 2.1. Histogram of Dataset Variable Types.

Regarding data types, the dataset contains a mixture of numerical, categorical, and boolean variables. Categorical variables represent the largest proportion of attributes, reflecting the descriptive nature of the dataset, while numerical variables capture wealth, demographic, and economic indicators. This

combination of variable types supports both descriptive analysis and predictive modeling. Observe Figure 2.1 to see more detail.

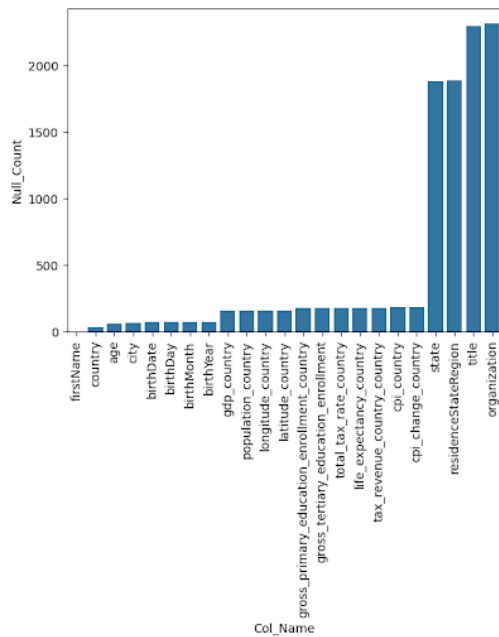


Figure 2.2. Histogram of null values on the dataset.

Finally, an analysis of missing values was performed to assess data quality before conducting the exploratory analysis. As shown in Figure 2.2, most variables contain little or no missing information. However, four attributes exhibit a substantially higher number of null values: organization, title, residenceStateRegion, and state. In particular, the variables organization and title contain missing values in the majority of records, suggesting that this information was either unavailable or not consistently reported across sources. These findings were later considered during the preprocessing stage, where highly incomplete attributes were evaluated for removal or imputation. Overall, the dataset presents a relatively good level of completeness, with missing values concentrated in a small subset of variables rather than being widespread across the entire dataset.

B) Analysis of Categorical variables

The dataset contains 18 categorical variables describing demographic, geographic, and wealth-related characteristics of billionaires. The exploratory analysis focused on identifying the most relevant patterns regarding wealth generation, geographic distribution, and demographic composition.

The analysis of billionaire industries revealed a strong concentration of wealth in a limited number of sectors. As shown in Figure 2.3, the largest groups correspond to Manufacturing, Technology, Finance & Investments, and Fashion & Retail. These sectors consistently appear among the most common industries associated with billionaires, suggesting that large-scale production, financial markets, and technological innovation remain the primary drivers of extreme wealth creation.

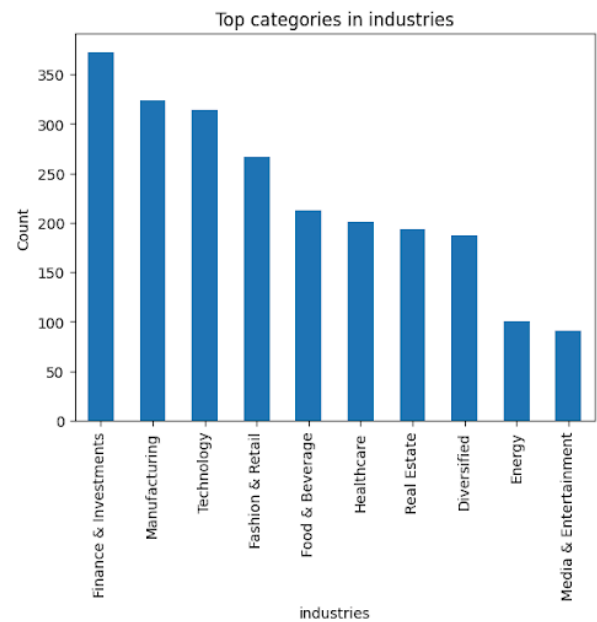


Figure 2.3. Histogram showing the distribution of the most common industries among billionaires

A similar pattern was observed when analyzing the geographic distribution of billionaires. The United States contains by far the largest number of billionaires, followed by China and, at a considerable distance, India. This concentration may be explained by the size of these economies, their large populations, and the presence of strong entrepreneurial and investment ecosystems that facilitate wealth creation.

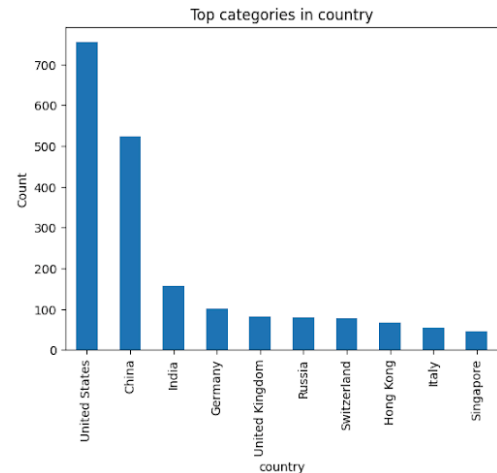


Figure 2.4. Distribution of the top 10 billionaire countries.

Finally, the demographic analysis revealed two important patterns. First, most billionaires are self-made rather than inheritors of wealth, indicating that entrepreneurship and business creation play a significant role in the accumulation of large fortunes.

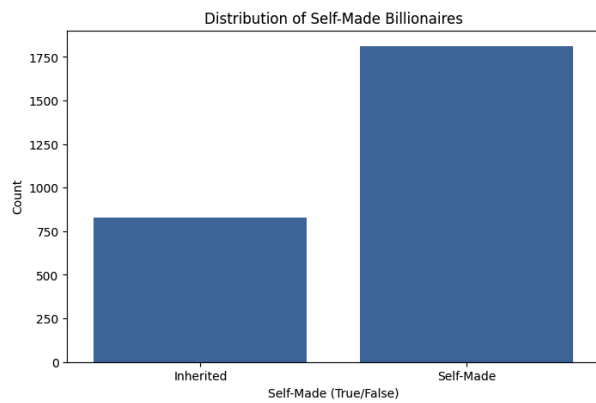


Figure 2.5. Distribution of self-made status

C) Analysis of Numerical variables

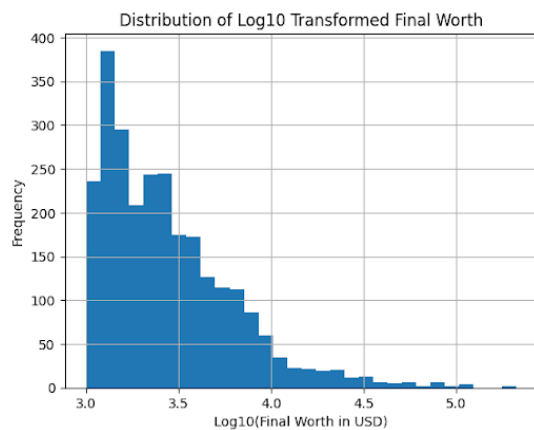


Figure 2.6. Histogram of the log₁₀-transformed final worth distribution among billionaires.

The numerical variables describe both individual billionaire characteristics and country-level socioeconomic indicators. The analysis focused on understanding the distribution of wealth, demographic patterns, and the economic environments associated with billionaire populations.

The distribution of net worth (finalWorth) presents a strong right-skewed behavior. As shown in Figure 2.6, most billionaires are concentrated within lower wealth ranges, while a small number of individuals possess substantially larger fortunes. To facilitate visualization, a logarithmic transformation was applied, reducing the impact of extreme values while preserving the overall distribution pattern. This result highlights the high inequality that exists even within the billionaire population itself.

The age distribution reveals that billionaires are predominantly older individuals, with most observations concentrated between approximately 50 and 75 years of age. The average age is around 65 years, suggesting that the accumulation of extreme wealth typically requires long-term business development, investment activity, and capital growth. Although a few younger billionaires exist, they represent uncommon cases compared to the overall

population.

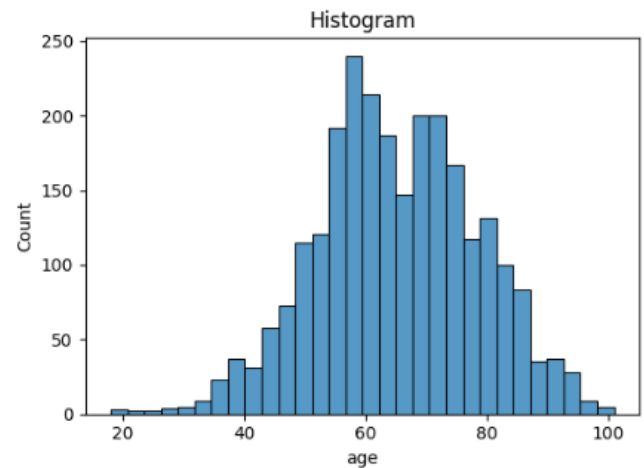


Figure 2.7. Histogram of the log₁₀-transformed final worth distribution among billionaires.

Country-level socioeconomic indicators such as GDP, education enrollment, life expectancy, taxation, and population exhibit considerable variability across countries. Large economies such as the United States and China appear as clear outliers in terms of GDP and population, while education and life expectancy show more stable distributions. Overall, these variables provide important contextual information for understanding the national environments in which billionaire wealth is generated and accumulated.

D) Correlation between variables

The correlation matrix reveals several meaningful relationships among the country-level variables. The strongest positive correlation is observed between GDP and population (0.61), indicating that larger countries tend to have greater economic output. Additionally, life expectancy shows a moderate positive correlation with tertiary education enrollment (0.58), suggesting that countries with higher levels of education generally achieve better health outcomes. Inflation-related indicators present notable relationships, with CPI change positively correlated with total tax rate (0.53) and CPI negatively correlated with life expectancy (-0.65). Overall, the matrix suggests that economic size, education, and health indicators tend to be interconnected, while inflation variables exhibit more independent dynamics from the remaining socioeconomic factors.

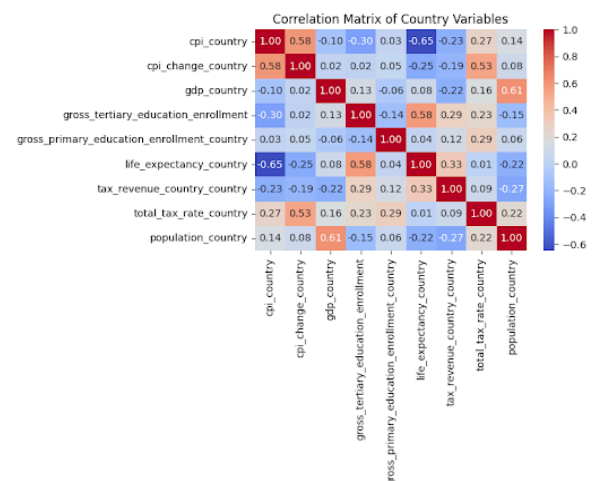


Figure 2.8 . Correlation Matrix of Country-Level Variables.

III. ASSOCIATION TASK

A) Data Preprocessing and Transactional Information

The association analysis aimed to identify structural relationships between billionaire wealth-generation patterns and national socioeconomic indicators. Since association rule mining requires transactional data, the original dataset underwent an extensive preprocessing pipeline involving aggregation, feature engineering, discretization, and binary transformation. Continuous variables related to wealth structure, macroeconomic indicators, education, taxation, inflation, and life expectancy were converted into categorical intervals using quantile-based binning and domain-specific thresholds. Additional indicators such as wealth concentration, sector diversification, self-made wealth ratios, wealth per capita, and billionaire wealth relative to GDP were also constructed.

After preprocessing, all categorical variables were transformed into a binary transactional representation through One-Hot Encoding, enabling compatibility with association rule mining algorithms. Missing values were handled using categorical placeholders such as unknown or 0 to ensure dataset consistency.

B) Apriori Algorithm

The Apriori algorithm was implemented to discover frequent itemsets and association rules linking wealth-generation structures with indicators of national well-being. A minimum support threshold of 0.1 was applied, while association rules were filtered using lift values greater than 1 to ensure positive statistical associations. To improve interpretability, antecedents were restricted to wealth-related variables and consequents to macroeconomic or social indicators such as inflation stability, education enrollment, taxation, and life expectancy.

The Apriori analysis revealed several recurring structural patterns. Frequent itemsets indicated that many countries in the dataset were characterized by low numbers of billionaires, limited economic diversification, and moderate inflation conditions. Most association rules exhibited support values between 10% and 15%, with confidence levels frequently ranging from 0.70 to 0.90, suggesting relatively strong predictive reliability.

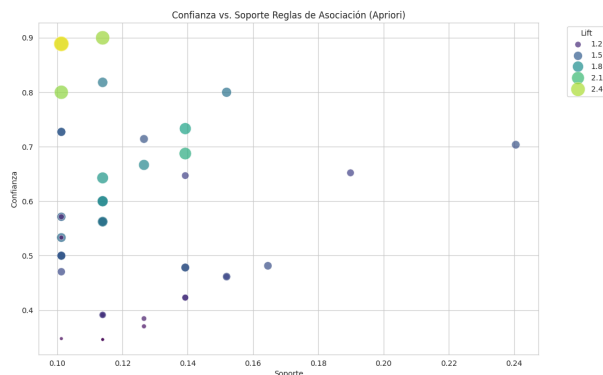


Figure 3.1. Confidence vs Support of Association Rules found by Apriori algorithm.

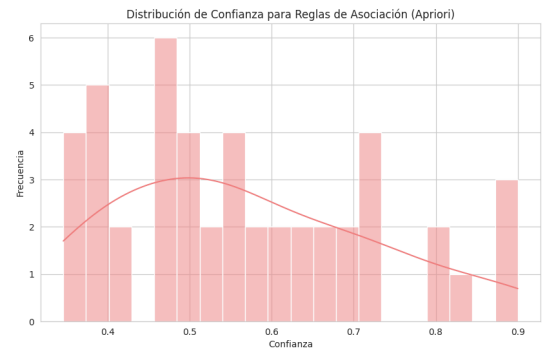


Figure 3.2. Confidence Distribution for Association Rules found by Aprior algorithm.

Among the most relevant rules, countries with low industrial and manufacturing participation were strongly associated with stable inflation conditions, reaching confidence values up to 0.90 and lift values above 2.3. Similarly, lower industrial concentration was repeatedly associated with higher life expectancy and improved macroeconomic stability. Another important relationship showed that countries with high participation in technology and heterogeneous “other” sectors frequently exhibited low tertiary education enrollment, suggesting that technological wealth concentration does not necessarily translate into broad educational development. Additionally, consumer discretionary sectors were positively associated with high life expectancy, while financial-sector participation and self-made wealth structures were linked to moderate inflation environments and specific taxation patterns.

C) FP-Growth Algorithm

The FP-Growth algorithm was also implemented using the same transactional dataset. Unlike Apriori, FP-Growth avoids explicit candidate generation by relying on the FP-Tree structure, improving computational efficiency while preserving the quality of discovered patterns. The algorithm generated 47 refined association rules with lift values extending up to approximately 2.75, confirming the existence of strong structural relationships between wealth accumulation patterns and socioeconomic indicators.

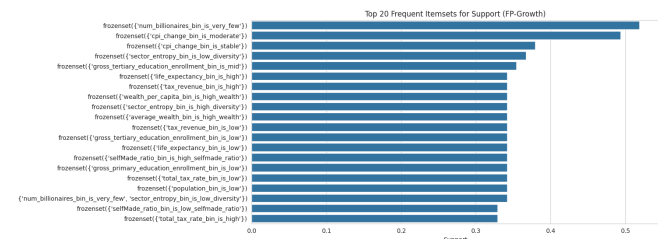


Figure 3.3. Top-20 Frequent Itemsets found by FP-Growth algorithm.

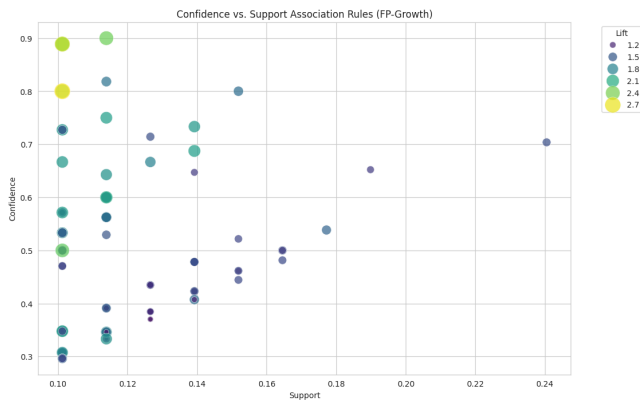


Figure 3.4. Confidence vs Support of Association Rules found by FP-Growth algorithm.

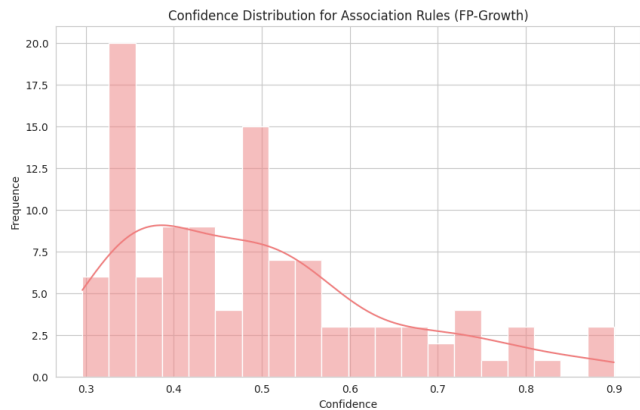


Figure 3.5. Confidence Distribution for Association Rules found by FP-Growth algorithm.

The strongest FP-Growth rules reinforced the relationship between low industrial dependence, stable inflation, and high life expectancy. Countries with reduced manufacturing concentration frequently displayed more favorable macroeconomic and social conditions, possibly reflecting diversified or service-oriented economic structures. High participation in “other” sectors was repeatedly associated with low tertiary education enrollment and, in some cases, lower life expectancy, suggesting weaker institutional or educational development in less specialized economies.

In contrast, countries with stronger consumer discretionary sectors tended to exhibit both inflation stability and higher life expectancy, indicating a relationship between mature consumer economies and improved living conditions. Overall, the FP-Growth results were highly consistent with those obtained through Apriori, validating the robustness of the identified structural associations.

D) Results

Both Apriori and FP-Growth successfully identified meaningful relationships between wealth-generation structures and indicators of national well-being. The results suggest that lower industrial dependence, diversified economic structures, and balanced entrepreneurial wealth generation are frequently associated with macroeconomic stability and stronger social indicators. Conversely, concentrated or poorly specialized wealth-generation patterns were often linked to weaker educational outcomes and less favorable socioeconomic conditions.

These findings demonstrate the usefulness of association rule mining for uncovering hidden structural relationships within complex socioeconomic datasets and provide insights into how

wealth concentration and industrial specialization may influence broader development indicators across countries.

Below, you will be able to find the most significant rules found by both algorithms:

TABLE 3.1
MOST SIGNIFICANT ASSOCIATION RULES IDENTIFIED BY APRIORI AND
FP-GROWTH ALGORITHMS

#	Structural Association Rule	Confidence	Lift	Main Interpretation
1	Low manufacturing and industrial participation → Stable CPI variation	0.90	2.37	Countries with lower industrial dependence tend to exhibit greater macroeconomic stability and lower inflation volatility.
2	Low industrial concentration → Stable inflation and high life expectancy	0.80	2.75	Diversified or service-oriented economies are frequently associated with both economic stability and favorable social conditions.
3	High technology and “other” sector participation → Low tertiary education enrollment	0.89	2.60	Technological wealth concentration does not necessarily imply broad educational development or inclusive human capital growth.
4	High participation of “other” sectors → Low tertiary education enrollment	0.69	2.01	Economies dominated by heterogeneous or weakly specialized sectors tend to exhibit weaker higher-education indicators.
5	Consumer discretionary sector concentration → High life expectancy	0.64	1.88	Consumer-oriented economies are commonly associated with higher living standards and stronger healthcare conditions.
6	Low industrial dependence → High life expectancy	0.80	2.34	Reduced manufacturing concentration is frequently associated with improved social and health outcomes.
7	Moderate self-made wealth and low industrialization → Stable inflation	0.75	1.97	Balanced entrepreneurial economies with lower industrial dependence tend to maintain more stable macroeconomic conditions.

#	Structural Association Rule	Confidence	Lift	Main Interpretation
8	High financial-sector participation → Moderate inflation	0.82	1.66	Economies driven by finance and investment activity often exhibit controlled but persistent inflationary behavior.
9	High “other” sector participation → Low life expectancy and low education enrollment	0.50	2.32	Weakly specialized economic structures may be associated with lower institutional development and poorer social indicators.
10	Consumer discretionary sectors → Stable inflation and high life expectancy	0.57	1.96	Mature consumer economies frequently combine macroeconomic stability with favorable quality-of-life indicators.

IV. CLUSTERING TASK

A) K-Means clustering and Structural wealth ecosystem

Clustering techniques were applied to identify structural wealth-generation patterns among countries. The goal was to determine whether countries naturally group into distinct wealth ecosystems, such as innovation-driven, finance-oriented, or resource-dependent economies.

Country-Level Feature Engineering

Since the dataset contains billionaire-level observations, country-level features were created by aggregating billionaire data by country. Countries with fewer than three billionaires were excluded to improve reliability.

The engineered features included sectoral proportions (e.g., technology, finance, and energy), the ratio of self-made billionaires, and the Herfindahl-Hirschman Index (HHI) to measure industrial concentration. In addition, socioeconomic indicators such as tertiary education enrollment, life expectancy, tax revenue, GDP, and population were incorporated to provide broader economic context.

TABLE 4.1
SUMMARY OF ENGINEERED FEATURES FOR COUNTRY

Feature	Description
tech_ratio	Proportion of billionaires in technology
finance_ratio	Proportion in finance
energy_ratio	Proportion in energy/mining
self_made_ratio	Ratio of self-made billionaires
hhi_index	Industrial concentration
billionaire_count	Number of billionaires

Feature	Description
tertiary_education	Education enrollment
life_expectancy	Average life expectancy

Before clustering, all numerical features were standardized using z-score normalization to ensure comparable scales across variables. PCA was then applied to reduce dimensionality and project the data into two principal components, capturing most of the variance and enabling visualization of similarities between countries.

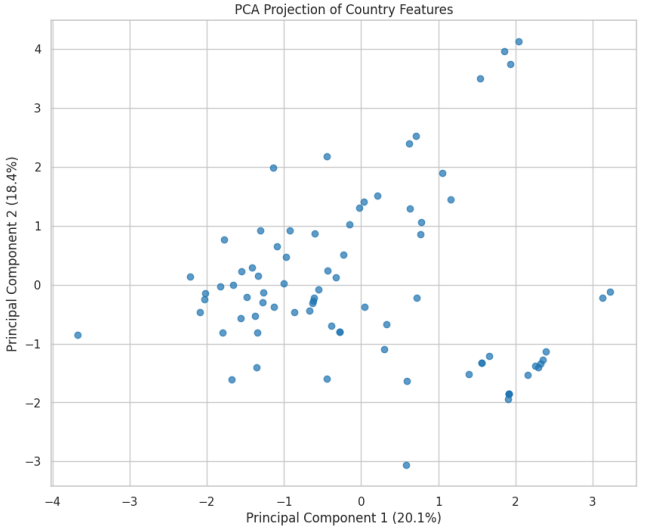


Figure 4.1. PCA Projection of Countries Based on Structural Wealth Features.

The Elbow Method and Silhouette Score were used to evaluate clustering performance. Both approaches indicated $k = 3$ as the most appropriate number of clusters, supporting the study’s theoretical framework.

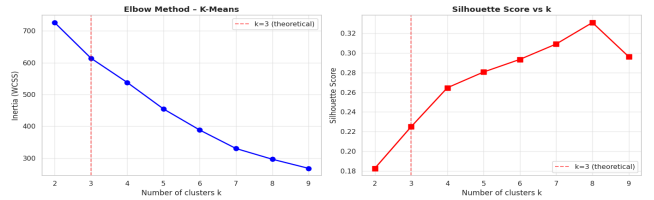


Figure 4.2. Elbow Method and Silhouette metrics.

K-Means was selected as the primary clustering method due to its interpretability and consistency with the hypothesis of three structural wealth ecosystems. The algorithm groups countries by minimizing within-cluster variance.

The normalized heatmap highlights differences in structural characteristics across clusters, revealing three distinct wealth ecosystems: innovation-oriented, resource-dependent, and finance-dominated economies.



Figure 4.3. K-Means Clustering Results Projected onto the PCA Space.

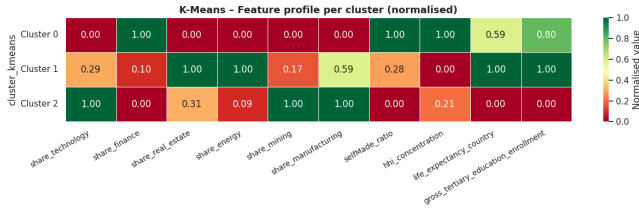


Figure 4.4. Heatmap of Normalized Cluster Profiles.

B) Density-Based and Hierarchical Clustering Approaches

DBSCAN and Hierarchical Clustering were applied to validate the K-Means solution. DBSCAN detects outlier countries, while Hierarchical Clustering reveals similarity relationships between countries, providing additional insight into global wealth-generation ecosystems.

DBSCAN Clustering

DBSCAN was applied to identify clusters and outlier countries without specifying the number of clusters in advance. The parameters used were $\epsilon = 2.5$ and $\text{min_samples} = 3$, with ϵ selected through a k-distance plot.

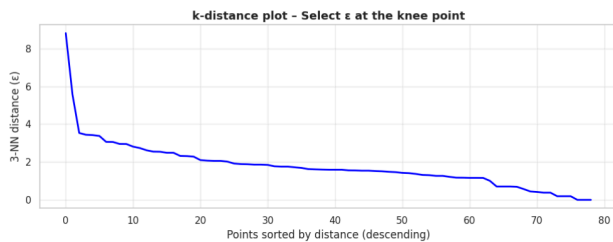


Figure 4.5. K-distance plot used to select ϵ for DBSCAN

DBSCAN identified multiple clusters and outlier countries, revealing atypical wealth-generation structures. The PCA projection shows the spatial distribution of these groups.

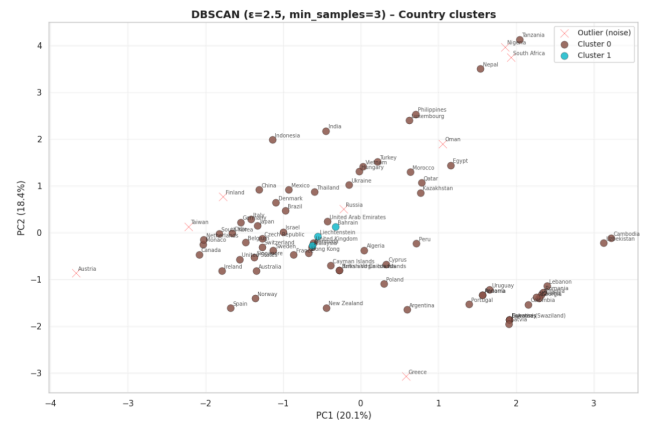


Figure 4.6. PCA visualization of DBSCAN clusters and outlier countries.

Outlier countries showed high sectoral concentration and lower diversification, demonstrating DBSCAN's effectiveness in identifying atypical economic structures.

Hierarchical Clustering using Ward Linkage

Hierarchical Clustering was applied using Ward linkage, which minimizes within-cluster variance during the merging process. The resulting dendrogram provides a visual and interpretable representation of similarity relationships among countries.

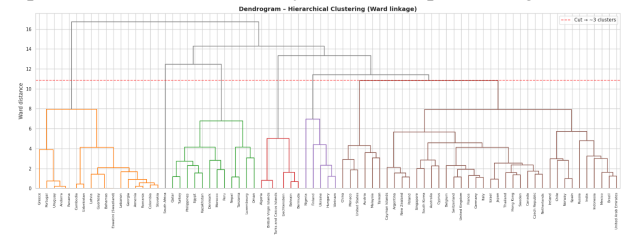


Figure 4.7. Dendrogram generated using Ward linkage.

The dendrogram suggested the presence of approximately three major clusters, supporting the proposed theoretical framework. Cluster quality was evaluated using the Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index, showing moderate agreement with the K-Means solution. The Adjusted Rand Index ($\text{ARI} \approx 0.69$) further indicated moderate similarity between the two methods.

Comparison of Clustering Algorithms

The three clustering methods provided complementary insights into global wealth-generation ecosystems. K-Means best matched the proposed three-archetype framework, DBSCAN identified atypical countries, and Hierarchical Clustering offered an interpretable view of country relationships through the dendrogram. Despite relatively low Silhouette Scores, all methods revealed meaningful patterns related to innovation, financial concentration, and resource dependence.

=== COMPARISON OF ALL THREE CLUSTERING ALGORITHMS ===

Algorithm	k / clusters found	Outliers detected	Silhouette ↑	Davies-Bouldin ↓	Calinski-Harabasz ↑	Silhouette ≥ 0.50	DBI < 1.00
K-Means	3	0	0.2249	1.6968	15.75	X	X
DBSCAN	2	8	0.2401	0.8253	6.16	X	✓
Hierarchical (Ward)	3	0	0.2278	1.5881	14.65	X	X

Figure 4.8. Comparison of clustering evaluation metrics

The classification task aimed to predict whether a billionaire is self-made or inherited their wealth. The target variable was therefore defined as **selfMade**, representing a binary classification problem.

A) Data Preparation

The classification dataset was obtained from the preprocessing stage described previously. Several attributes considered irrelevant or highly specific to individual identities were removed, including personal names, cities, birth dates, and source descriptions. Additional feature engineering was performed on the title variable, generating binary indicators that capture important leadership characteristics such as founder status, CEO role, chairman position, and investor profile.

The final feature set included demographic attributes, industry information, country-level socioeconomic indicators, and engineered binary variables. Numerical variables were normalized using Min-Max scaling, while categorical attributes were transformed through One-Hot Encoding. This preprocessing ensured compatibility with machine learning algorithms while preserving the information contained in the original dataset.

B) Decision Tree Classification

A Decision Tree classifier was implemented to predict whether a billionaire is self-made or inherited their wealth. The model was trained using the entropy criterion, which follows the same information-theoretic principles employed by the C4.5 algorithm. To reduce overfitting and improve generalization, several regularization parameters were introduced, including a maximum tree depth of 20, a minimum of 30 samples per leaf node, and a minimum of 60 samples required to perform a split.

Before training, numerical variables were normalized using Min-Max scaling, while categorical variables were transformed through One-Hot Encoding. The final feature set included demographic information, industry affiliation, country-level socioeconomic indicators, and engineered binary variables derived from leadership positions such as CEO, chairman, and investor status.

The dataset was divided into training and testing subsets, resulting in 1,846 observations for training and 396 observations for testing. The target variable presented a moderate class imbalance, with self-made billionaires representing the majority class.

The Decision Tree classifier achieved an overall accuracy of 94% on the test dataset, demonstrating strong predictive performance. For the self-made billionaire class, the model obtained a precision of 0.92, a recall of 0.99, and an F1-score of 0.96. These results indicate that the classifier successfully identified nearly all self-made billionaires, making very few false negative predictions.

For the inherited wealth class, the model achieved a precision of 0.98, a recall of 0.82, and an F1-score of 0.89. The high precision suggests that when the model predicts inherited wealth, the prediction is highly reliable. However, the lower recall indicates that a portion of inherited-wealth individuals were incorrectly classified as self-made, which may be partially explained by the imbalance between classes.

Considering both categories simultaneously, the model achieved a macro-average precision of 0.95, a macro-average recall of 0.91, and a macro-average F1-score of 0.93. The weighted average

F1-score reached 0.94, confirming that the classifier maintained strong performance across the entire dataset.

These results demonstrate that demographic characteristics, industry information, citizenship, economic indicators, and leadership-related variables contain substantial predictive information for distinguishing between self-made and inherited billionaire wealth profiles. Furthermore, the Decision Tree offers an important advantage in terms of interpretability, as the generated structure can be visualized and analyzed to understand the decision rules used during classification.

Overall, the Decision Tree classifier provided accurate and interpretable predictions, establishing a strong baseline model for the billionaire wealth classification problem.

C) Naive Bayes Classification

As a second classification approach, a Gaussian Naive Bayes classifier was implemented to predict whether a billionaire is self-made or inherited their wealth. Naive Bayes is a probabilistic learning algorithm based on Bayes' Theorem and assumes conditional independence among predictor variables given the target class.

The same preprocessed dataset used in the Decision Tree experiment was employed. Numerical attributes were normalized using Min-Max scaling, categorical variables were transformed through One-Hot Encoding, and binary variables were kept unchanged. Unlike the Decision Tree model, the dataset was divided into training, validation, and testing subsets using a 70%-15%-15% split strategy. The target variable was encoded using Label Encoding before model training.

The Gaussian Naive Bayes classifier was trained on the training subset and evaluated on both validation and testing datasets. The results revealed substantially lower predictive performance compared to the Decision Tree classifier.

On the test dataset, the model achieved an overall accuracy of 37%. For the inherited wealth class (False), the classifier obtained a precision of 0.33, a recall of 0.96, and an F1-score of 0.49. In contrast, for the self-made billionaire class (True), the model achieved a precision of 0.83, a recall of only 0.09, and an F1-score of 0.17.

The macro-average precision, recall, and F1-score were 0.58, 0.53, and 0.33, respectively, while the weighted average F1-score reached only 0.27. Similar results were observed on the validation dataset, where the model achieved an accuracy of 36%, indicating that the poor performance was consistent rather than caused by overfitting.

These results suggest that the assumptions underlying the Gaussian Naive Bayes model are not well suited for this dataset. In particular, the independence assumption is likely violated because many of the predictor variables are naturally correlated. Previous exploratory analysis showed meaningful relationships among socioeconomic indicators such as GDP, education enrollment, life expectancy, and taxation variables. Furthermore, categorical variables describing industries, citizenship, and leadership roles may interact in complex ways that cannot be adequately captured by a Naive Bayes classifier.

Another notable issue is the extremely low recall obtained for the self-made billionaire class. Although the model achieved relatively high precision for this category, it failed to identify the majority of self-made billionaires, leading to a large number of false negative predictions. This behavior indicates that the classifier was unable to learn a sufficiently discriminative representation of the underlying class structure.

Overall, the Gaussian Naive Bayes classifier demonstrated limited effectiveness for this classification problem. The results highlight the importance of considering dependencies among variables and suggest that more flexible algorithms, such as Decision Trees and Neural Networks, are better suited for modeling the complex relationships present in billionaire wealth data.

D) Neural Network Classification

To further investigate the classification problem, Artificial Neural Networks were employed as an alternative to traditional machine learning algorithms. Two neural network architectures were evaluated: a Multilayer Perceptron (MLP) with regularization techniques and a TabNet model specifically designed for tabular data.

Prior to training, the dataset underwent an extensive preprocessing phase. Categorical variables were transformed through One-Hot Encoding, low-frequency citizenship categories were grouped into an "Other" class to reduce dimensionality, and numerical features were standardized using the StandardScaler technique. After preprocessing, the final dataset contained 77 input features.

The data were divided into training, validation, and testing subsets using a 70%-15%-15% split strategy. The resulting partitions consisted of 1,846 training samples, 396 validation samples, and 396 testing samples.

The Multilayer Perceptron architecture was composed of two hidden layers containing 128 and 64 neurons, respectively. ReLU activation functions were adopted for both hidden layers, while a Softmax activation function was used in the output layer to perform binary classification. To improve generalization and reduce overfitting, L2 regularization, Batch Normalization, and Dropout layers with a dropout rate of 0.40 were incorporated into the architecture. Model optimization was performed using the Adam optimizer with a learning rate of 0.007.

The network was trained for 50 epochs using a batch size of 32. Throughout the training process, both training and validation accuracies remained stable and converged rapidly, indicating effective learning behavior and limited overfitting. Validation accuracy consistently remained above 94%, reaching peaks close to 96% during training.

When evaluated on the test dataset, the MLP achieved a test accuracy of 94.19% with a corresponding loss value of 0.1722. These results demonstrate that the neural network successfully captured the complex nonlinear relationships among demographic, industrial, and socioeconomic variables associated with billionaire wealth profiles.

To provide an additional benchmark, a TabNet classifier was also trained using the same dataset. TabNet combines sequential attention mechanisms with feature selection strategies specifically tailored for tabular data. The model achieved a validation AUC of 0.972 and a final test accuracy of 92.68%.

The classification report obtained for TabNet showed strong performance across both classes. For the inherited wealth class, the model achieved a precision of 0.87, a recall of 0.90, and an F1-score of 0.89. For the self-made billionaire class, the precision reached 0.96, the recall 0.94, and the F1-score 0.95. The overall weighted F1-score was 0.93.

Comparing both neural network approaches, the Multilayer Perceptron achieved the highest overall classification accuracy and demonstrated excellent stability throughout training. These findings suggest that deep learning models are capable of effectively capturing the complex interactions among

wealth-related predictors and represent a highly competitive solution for the billionaire self-made classification task.

VI. CONCLUSIONS

This project demonstrated the effectiveness of data mining techniques for uncovering meaningful patterns related to global wealth generation and distribution. Through the analysis of a billionaire dataset containing individual and country-level socioeconomic information, it was possible to identify structural relationships between wealth accumulation, economic environments, and social development indicators.

The exploratory analysis revealed a strong concentration of billionaire wealth in a limited number of industries, particularly Manufacturing, Technology, and Finance & Investments, as well as a significant geographic concentration in countries such as the United States and China. Additionally, the results showed that most billionaires are self-made, highlighting the importance of entrepreneurship and business creation in the accumulation of extreme wealth. The association analysis, conducted through both Apriori and FP-Growth algorithms, successfully identified recurring relationships between wealth-generation structures and national socioeconomic conditions. The findings suggest that countries with lower industrial dependence and more diversified economic structures tend to exhibit greater macroeconomic stability, higher life expectancy, and more favorable social outcomes. The consistency of the results obtained by both algorithms reinforces the reliability of the discovered patterns.

The clustering task revealed the existence of distinct national wealth ecosystems. K-Means, DBSCAN, and Hierarchical Clustering consistently identified groups of countries characterized by different combinations of innovation, financial specialization, and resource dependence. These results demonstrate that billionaire wealth structures are strongly influenced by broader economic and institutional contexts and that countries naturally organize into identifiable wealth-generation archetypes. Regarding classification, the study confirmed that billionaire profiles can be accurately predicted using demographic, industrial, and socioeconomic variables. The Decision Tree and Neural Network models achieved excellent predictive performance, with accuracies above 94%, while Naive Bayes performed considerably worse due to the strong dependencies among the predictor variables. Among all evaluated models, the Multilayer Perceptron achieved the best overall performance, demonstrating the ability of deep learning techniques to capture complex nonlinear relationships within tabular socioeconomic data.

Overall, the project highlights the value of combining data preprocessing, feature engineering, association mining, clustering, and classification techniques to extract actionable insights from complex socioeconomic datasets. The findings contribute to a better understanding of the structural factors associated with extreme wealth creation and provide a data-driven perspective on how economic specialization, entrepreneurship, and national development indicators interact in shaping billionaire wealth ecosystems worldwide.

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